Spot Micro Robot Dog

Updated parts list | 23 Oct 2026

1. Computing & Sensors

Item	Qty	Notes
Raspberry Pi 3B	1	Main computer.
microSD card	1	16-32 GB.
Raspberry Pi Camera v2 + ribbon cable	1	Cable length 20-30 cm. The plate has no camera mount, so a separate 3D-printed mount is needed, plus 4x M2x6 screws with nuts.
MPU-6050 (IMU)	1	Motion/orientation sensor. Stays in the build.
LCD 16x2 with I2C	1	Small status display (I2C address 0x27).
Dupont jumper wires (F-F)	1 set	For the I2C connections between the Pi, PCA9685 and LCD.

2. Drive & Mechanics

Item	Qty	Notes
PCA9685 servo driver	1	16-channel PWM board that controls the servos.
PDI-HV5523MG servos	12	High-voltage metal-gear servos (3 per leg).
3-pin servo extension cables	12	One per servo.
Spot Micro 3D-printed parts	1 set	The frame and leg parts.
Fasteners	-	See the fastener table below.
F625zz bearings	8	Also listed in the fastener table.

3. Power

Item	Qty	Notes
2S LiPo battery, 4000 mAh	1	7.4 V nominal.
UBEC 5V / 5A	1	Steps battery voltage down to 5 V for the Pi and electronics.
Rocker switch, rated 20 A or more	1	Main power on/off. Must handle at least the fuse current (15-20 A).
XT60 connector and Y-splitter	1 set	Splits battery power between the servos and the UBEC.
Fuse, 15-20 A	1	Protects against short circuits.
Capacitor, 1000-2200 uF, 16 V or higher	1	Smooths voltage spikes from the servos.
Wire, 14-16 AWG	-	For the power lines.
Balance charger	1	For safely charging the LiPo.
Low-voltage buzzer	1	Alarm when the battery is running low.

4. Voice

Item	Qty	Notes
USB microphone	1	Voice input.
USB speaker	1	Voice output. Alternative: USB sound card + passive speaker + amplifier.

5. Optional

Item	Qty	Notes
ADS1115 ADC	1	Used to measure battery voltage.
Resistors for voltage divider	2	For the ADS1115: the battery (up to 8.4 V) must not go to the ADC input directly. E.g. 10 kOhm + 10 kOhm halves it to 4.2 V max.

6. Fasteners & Bearings

Item	Qty	Notes
Bolt M5x15	8	
Bolt M4x20	40	
Bolt M4x15	8	
Nut M4	48	Matches the total M4 bolts (40 + 8).
Bolt M3x20	4	
Bolt M3x10	26	
Nut M3	14	
Bolt M2.5x10	4	For the Raspberry Pi board.
Nut M2.5	4	For the Raspberry Pi board.
Bearing F625zz	8	

Note: the 4x M2x6 screws with nuts for the camera mount (section 1) are not in this table; order them separately.